

Solution Requirements (Functional & Non-functional)

1. Functional Requirements

The functional requirements describe the core features and capabilities of the Flood Prediction System.

Requirement ID	Functional Requirement
FR1	The system shall allow users to enter weather parameters such as annual rainfall, seasonal rainfall, and cloud visibility.
FR2	The system shall validate the entered input values before processing.
FR3	The system shall preprocess the input data to match the format required by the machine learning model.
FR4	The system shall use the trained machine learning model (XGBoost/Random Forest/Decision Tree/KNN) to predict flood occurrence.
FR5	The system shall display the prediction result as "Flood" or "No Flood" along with the prediction confidence (if available).
FR6	The system shall allow multiple predictions without restarting the application.
FR7	The system shall load the saved machine learning model automatically when the Flask application starts.
FR8	The system shall provide a simple and user-friendly web interface for disaster management authorities and users.

2. Non-Functional Requirements

The non-functional requirements define the quality attributes and operational characteristics of the system.

Requirement ID	Non-Functional Requirement
NFR1	The system should provide prediction results within a few seconds.
NFR2	The application should maintain high prediction accuracy (approximately 96.55% using the XGBoost model).
NFR3	The web interface should be responsive and easy to use.
NFR4	The system should be available 24×7 when deployed on the cloud.
NFR5	The application should handle invalid or missing input values gracefully by displaying appropriate error messages.
NFR6	The system should be scalable to support multiple users simultaneously.
NFR7	The application should ensure secure handling of user requests and prevent unauthorized access.
NFR8	The system should be maintainable, allowing future updates such as integrating real-time weather APIs or IoT sensors.
NFR9	The application should be compatible with modern web browsers and mobile devices.
NFR10	The system should be reliable and produce consistent predictions for the same input data.

Description

The Flood Prediction System is designed to provide reliable and accurate flood risk predictions based on weather parameters. The functional requirements define the system's primary operations, including data input, preprocessing, prediction using machine learning algorithms, and displaying results through a Flask-based web application. These features enable users to assess flood risks efficiently.

The non-functional requirements focus on the system's overall quality, including performance, accuracy, reliability, scalability, security, usability, and maintainability. Together, these requirements ensure that the solution is suitable for real-world disaster management, where timely and dependable flood predictions are essential for effective decision-making.